CarCheck Agent — Design System

> Companion to the *CarCheck Agent PRD v1.0* and `architecture.md`.
> Direction: **light backgrounds, pastel colours, evidence first.** Mobile-first React PWA.

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1. Design intent

Who it's for. A first-time used-car buyer standing next to a car, phone in hand, nervous about being fooled.

Primary job. Help them film a good walkaround, then understand what the report says and how sure it is.

Feeling. Calm, clear and honest. A friendly second opinion, not an alarm panel and not a verdict. The PRD says CarCheck is *screening, not certification*, so the visual language must never feel like a stamp of approval or a warning siren.

The one memorable thing. The **confidence band**: a soft, segmented bar that sits on every finding and shows how sure the system is. Uncertainty is part of the look, not hidden in fine print. Everything else stays quiet so this can be seen.

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2. Principles

1. **Never a score without its evidence.** The score ring always sits above, or beside, the findings that produced it.
2. **Make uncertainty visible.** Confidence is drawn, not just printed. *Uncertain* findings look different from confident ones.
3. **Pastel means gentle, not vague.** Pastel fills carry the mood; dark ink carries the meaning. Text is never pastel.
4. **Colour is never the only signal.** Every colour state pairs with an icon and a word.
5. **Plain words.** "Possible repaint", not "Delta-E exceedance". Sentence case everywhere.
6. **Spend boldness in one place.** The confidence band and the score ring get character. Cards, lists and chrome stay restrained.
7. **Nothing accusatory.** Findings are "possible", always with a next step ("Ask the seller…", "Have a mechanic check…").

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3. Colour

3.1 Backgrounds (all light)

Token	Hex	Use
`--bg-canvas`	`#F6F9FC`	App background (a very pale, cool blue-white)

Token	Hex	Use
`--bg-surface`	`#FFFFFF`	Cards, sheets, report pages
`--bg-sunken`	`#EDF2F8`	Inset areas, image wells, input fields
`--bg-overlay`	`rgba(255,255,255,0.82)`	Chrome floating over the camera feed (with backdrop blur)

3.2 Pastel palette

Each pastel has three tones: a **fill** for backgrounds, a **line** for borders and fills of graphics, and an **ink** for text and icons placed on that fill.

Name	Fill	Line	Ink	Role
Sky	`#DCEBFA`	`#9CC5EE`	`#1D4E89`	Primary actions, links, progress, neutral info
Mint	`#D9F2E6`	`#8FD1B0`	`#14513A`	Looks fine, agent agrees, captured

Name	Fill	Line	Ink	Role
Butter	`#FBF0C9`	`#EBD27A`	`#6B4E00`	Uncertain, check this, caution
Peach	`#FCE0D6`	`#F1A98F`	`#8A3A1F`	Possible issue, moderate severity
Rose	`#F9DCE4`	`#E8A0B5`	`#8A2845`	Major severity, significant concern
Lavender	`#E8E3F8`	`#B9ABE8`	`#4B3A8C`	The agent's voice: notes, summaries, questions

3.3 Text and structure

Token	Hex	Use
`--ink-900`	`#1F2A37`	Headings, primary text
`--ink-700`	`#3D4B5C`	Body text
`--ink-500`	`#5F6E80`	Secondary text, captions
`--ink-300`	`#9AA7B6`	Disabled text only (not for content)

Token	Hex	Use
`--line-subtle`	`#E1E8F0`	Dividers, card outlines
`--line-strong`	`#C9D4E0`	Input borders, focused structure
`--focus-ring`	`#5B9BE0`	Keyboard focus (2px, 2px offset)

3.4 Semantic mapping

Severity (damage and findings)

Severity	Pastel	Icon	Label
Minor	Butter	small dot	Minor
Moderate	Peach	triangle	Moderate
Major	Rose	filled diamond	Major

Agent vision review

Verdict	Pastel	Icon	Label
Agree	Mint	check	Second opinion agrees
Uncertain	Butter	question mark	Second opinion unsure
Disagree	Peach	split arrows	Second opinion disagrees

Score bands (the score is a guide, not a grade; wording stays soft)

Score	Pastel	Headline
80–100	Mint	Looks sound in what we checked
60–79	Butter	A few things worth checking
40–59	Peach	Several concerns to follow up
0–39	Rose	Significant concerns found

No band says "safe", "clean" or "accident-free". Every band sits next to the coverage summary and the disclaimer.

3.5 Contrast

Ink-on-fill pairs above are chosen for strong contrast (targeting WCAG AA or better for normal text). Run them through a contrast checker once real values are final. Pastel **line** tones are decorative or graphical only, and any meaningful graphic also carries an icon or label.

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4. Typography

Role	Typeface	Why
Display and headings	**Bricolage Grotesque**	Friendly, slightly quirky grotesque with warmth; gives the product a personality without feeling clinical

Role	Typeface	Why
Body, UI, numbers	**Figtree**	Rounded, very legible at small sizes on phones; supports tabular numerals for measurements

Fallbacks: `system-ui, -apple-system, "Segoe UI", sans-serif`. Load with `font-display: swap`. For regional languages later, pair with Noto Sans of the relevant script.

Type scale (mobile, 16px base, ratio about 1.2)

Token	Size / line	Weight	Use
`display`	40 / 44	Bricolage 700	Score number
`h1`	28 / 34	Bricolage 700	Screen titles
`h2`	22 / 28	Bricolage 600	Section titles
`h3`	18 / 24	Figtree 600	Finding titles
`body`	16 / 24	Figtree 400	Default text
`body-sm`	14 / 20	Figtree 400	Captions, notes

Token	Size / line	Weight	Use
`caption`	12 / 16	Figtree 500	Timestamps, legal footnotes (never below 12)

Rules:

- Sentence case for all headings, buttons and labels. No all-caps labels or tracked-out eyebrows.
- Line length under about 70 characters for body text.
- Use `font-variant-numeric: tabular-nums` for gaps, percentages and scores so numbers don't jitter.
- Measurements always carry units and a range where the PRD requires it, e.g. "tread about 3–4 mm".

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5. Layout, shape and depth

Grid and spacing. 4px base unit. Spacing scale: 4, 8, 12, 16, 24, 32, 48. Screen padding 16px. Max content width 560px (phones and small tablets); the report widens to a two-column layout above 900px.

Touch targets. Minimum 48×48px (the PRD asks for large targets). Primary buttons 56px high.

Radius follows hierarchy (not one radius for everything):

Element	Radius
Chips and tags	999px (pill)

Element	Radius
Buttons, inputs	14px
Cards	20px
Bottom sheets	28px (top corners)
Evidence images	12px (a tighter, more document-like edge)

Depth. Mostly flat. Separation comes from fill and a 1px `--line-subtle` outline, not shadows.

- Resting cards: no shadow, outline only.
- Floating things (camera chrome, bottom sheet, toast): one soft tinted shadow, `0 8px 24px rgba(31,42,55,0.08)`.

Alignment. Left-aligned text throughout. Only the score ring and empty states are centred.

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6. Components

6.1 Buttons

Type	Fill	Text	Use
Primary	Sky fill `#DCEBFA`, 1px Sky line	Sky ink	One per screen: Start inspection, Upload, Retake

Type	Fill	Text	Use
Secondary	White, 1px `--line-strong`	`--ink-900`	Download PDF, Share
Quiet	None	Sky ink	Skip, Learn more
Destructive	White, 1px Rose line	Rose ink	Delete inspection (always confirms)

States: pressed darkens the fill about 6%; disabled uses `--bg-sunken` with `--ink-500`; focus shows the 2px focus ring. Button labels say exactly what happens: "Start inspection", "Retake rear-left", "Delete inspection and media". No arrows appended to labels.

6.2 Confidence band (signature component)

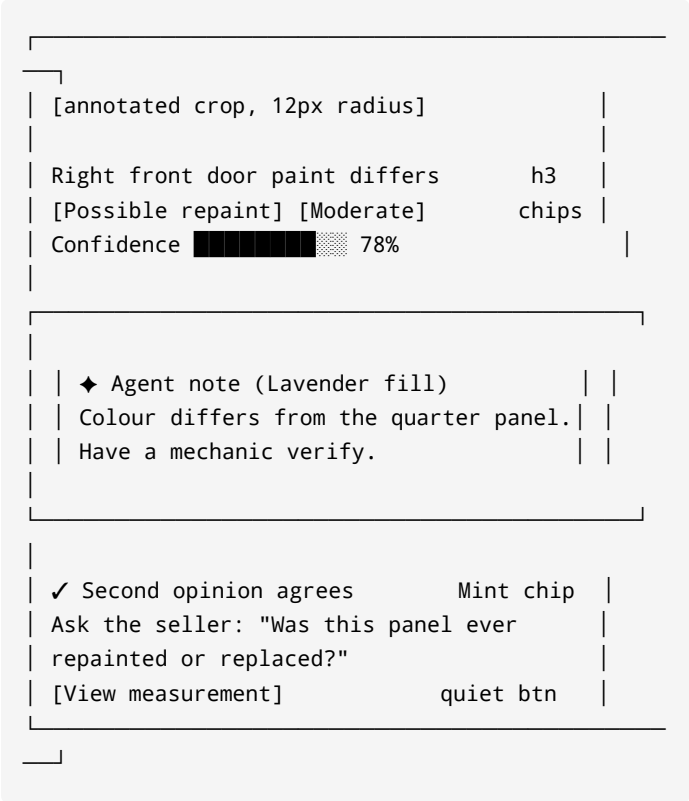
A horizontal bar made of 10 soft segments with the percentage beside it.

```
Confidence   ▓▓▓▓▓▓▓▓▓▓  78%
```

- Filled segments use the finding's severity pastel **line** tone; empty segments use `--bg-sunken`.
- **Uncertain** findings (vision review disagreed or unsure) switch to Butter, show a dashed outline around the band, and add the label "Uncertain".

- Always shown at the same position on every finding card so users learn where to look.
- Screen readers get: "Confidence 78 percent".

6.3 Finding card



- Evidence image first, because evidence leads.
- "View measurement" opens a sheet with the raw number (e.g. Delta-E value, gap in mm) and the frame it came from.

- Findings with the *uncertain* label get the dashed confidence band and a Butter chip; they are never hidden.

6.3b Annotations on evidence images

Photos vary, so annotation colours must be readable on real car paint.

- Bounding boxes and gap lines: 3px stroke in the severity pastel **line** tone, plus a 1px white outer halo and a 1px dark inner edge so they read on any background.
- Labels sit on a white pill with `--ink-900` text, positioned outside the box where possible.
- Plates and faces are blurred by default (a soft frosted blur, not a black box).

6.4 Score ring

- 160px ring, 14px stroke, rounded caps, drawn in the band's pastel line tone over a `--bg-sunken` track.
- Score number in `display` type at the centre; below it, the band headline ("A few things worth checking").
- Directly underneath: the coverage chip ("8 of 8 angles") and the line "Screening only, not a certification."
- If coverage is reduced, a Butter banner sits above the ring: "We couldn't check the rear-left corner."

6.5 Step indicator (upload and processing)

Four steps: Quality check, Analysis, Agent review, Report.

- Horizontal on mobile with a thin Sky-line track.
- Completed steps: Mint fill with a check. Current step: Sky fill with a gentle looping progress shimmer. Upcoming: `-`

`-bg-sunken`.

- Plain status text under it: "Checking each angle for blur and glare…".
- Numbers are used because this *is* a sequence.

6.6 Viewpoint progress ring (guided capture)

Eight arc segments around a small car silhouette, one per viewpoint (front, front-left, left, rear-left, rear, rear-right, right, front-right).

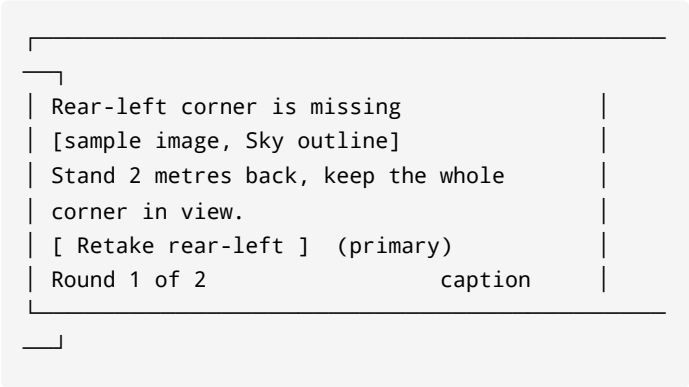
- Pending: `--line-strong`. Current: Sky line, slightly thicker. Captured: Mint line with a tick.
- Shown in the top corner over the camera, inside a `--bg-overlay` pill.

6.7 Capture overlay

Camera feed is the only dark area in the app, and that's fine: it's a live photo. All chrome on top stays light.

- **Ghost outline:** a 2px white outline of the car silhouette for the current viewpoint, with a soft 1px dark edge so it reads on bright and dark scenes.
- **Warnings** (glare, blur, moving too fast): a `--bg-overlay` pill at the top with a Butter fill and icon, e.g. "Too fast. Slow down a little." Warnings fade after 3 seconds if the issue clears. Never red, never blocking.
- **Prompt:** one short line at the bottom in an overlay card: "Stand at the front-left corner and move slowly right."
- **Record control:** 72px circular button, white with a Sky ring.

6.8 Retake request card



Round counter is shown so the user knows the retake loop is bounded. After round 2, the card is replaced by the Butter reduced-coverage banner.

6.9 Disclaimer banner

Always on the report, never collapsed.

- Lavender fill, 1px Lavender line, info icon.
- Text: "CarCheck is a screening tool, not a certification. Lighting, dirt and paint type can affect results. Have a mechanic verify anything flagged."
- Lives at the bottom of the report and as a short one-line version under the score ring.

6.10 Seller questions list

A checklist in Lavender-tinted rows. Each row: question text, a checkbox, and a link back to the finding that prompted it. Checkable so buyers can take it to the seller.

6.11 Chips

Pill shape, 28px high, pastel fill + matching ink + icon. Examples: `Possible repaint`, `Minor`, `Second opinion agrees`, `8 of 8 angles`.

6.12 Toasts and errors

- Toasts: white, floating shadow, Mint check for success. Same verb as the button ("Report deleted").
- Errors explain what happened and what to do, with no apology: "Upload stopped at 62%. Check your connection, then tap Resume."
- Error surfaces use Peach fill with Peach ink; never full-screen red.

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7. Screens (mobile wireframes)

7.1 Home and sign-in

```
┌──────────────────────────────────┐
│  CarCheck                        │
│                                  │
│  A second opinion on a           │
│  used car, from your             │
│  phone.                          │
│                                  │
│  [ car illustration,             │
│    pastel line art ]             │
│                                  │
│  [ Start inspection ]            │
│  Sign in                         │
│                                  │
```

```
│  Screening, not a        │
│  certification.          │
└──────────────────────────┘
```

7.2 Car details

Make, model, year (large inputs, `--bg-sunken` fields with `--line-strong` border), optional notes. Beneath: three tip rows with icons: "Film in daylight", "Wipe off dust and water", "Keep a printed marker or the number plate in view".

7.3 Guided capture

```
┌──────────────────────────┐
│ ◔ 3/8       [ Glare ]    │  ← viewpoint ring + warning pill
│                          │
│     (live camera feed)   │
│        ┌─────────┐       │
│        │ ghost │         │
│        └─────────┘       │
│                          │
│ Move slowly to the left  │
│ side of the car.         │
│          ( ● )           │
└──────────────────────────┘
```

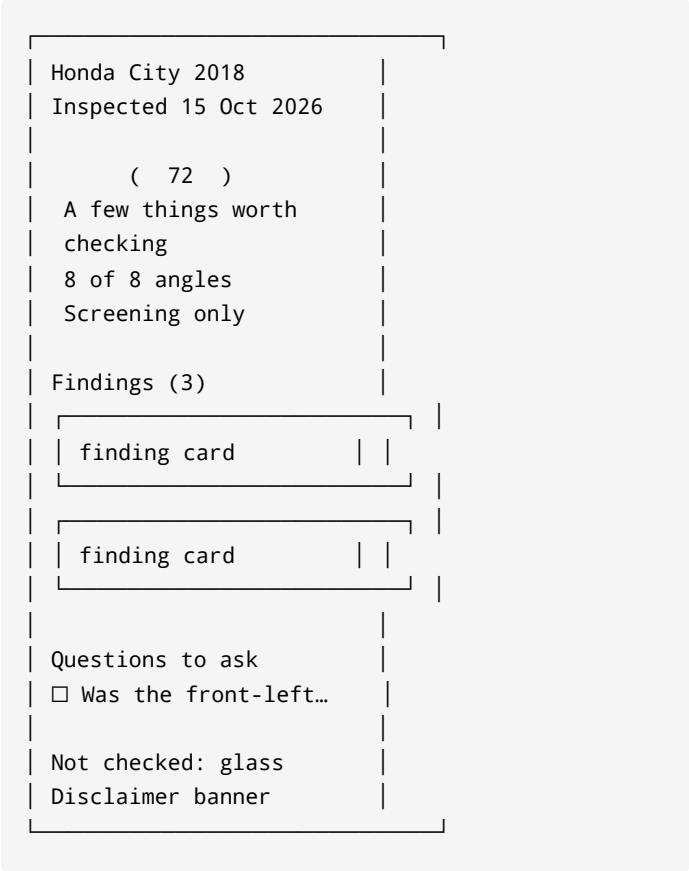
7.4 Upload and processing

Upload progress bar (Sky) with percentage, then the four-step indicator. A quiet Lavender card explains what's happening: "We measure the car with image analysis, then an AI reviewer double-checks anything unusual."

7.5 Retake request

Retake card (6.8) at the top, the coverage ring below showing captured and missing angles. Primary button returns straight to capture for that viewpoint only.

7.6 Report



Order matches the PRD report template: car details, score with one-line summary, coverage, findings, seller questions,

limitations and disclaimer. "Not checked" lists any analyses skipped so absence is never read as "fine".

7.7 Share and export

Download PDF, copy share link, revoke link, delete inspection. Delete uses the destructive button and a confirmation sheet that states what is removed.

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8. Report PDF

- White page, `--ink-900` text, same type pair; fits A4 and prints cleanly in greyscale because every status also has an icon and word.
- Pastel fills are kept as light tints; evidence images at full width with their annotations.
- Same section order as the in-app report; disclaimer on every page footer in `caption` size.

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9. Motion

- **One orchestrated moment:** the score ring draws itself once when the report first opens (about 900 ms, ease-out), with the number counting up in step.
- Everything else only responds to user action: sheets slide up, accordions expand, buttons press, checkboxes tick.
- Processing shimmer on the current step is the only looping animation.

- Respect `prefers-reduced-motion`: the ring appears already drawn; shimmer becomes a static highlight.

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10. Accessibility

- Contrast targets AA or better for all text; verify final values with a checker.
- Every colour state has an icon and a text label (works for colour-blind users and greyscale printing).
- Visible keyboard focus on every interactive element.
- Touch targets at least 48px.
- Images have alt text generated from the finding ("Close-up of right front door, area marked as possible repaint").
- Camera warnings are announced via an ARIA live region.
- Respect system text size up to 200% without breaking layout.
- Plan for right-to-left and regional-language support later: no text baked into images, strings come from a message catalogue.

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11. Voice and copy

Plain verbs, sentence case, active voice, no hype.

Situation	Write	Avoid
Start button	Start inspection	Begin your journey

Situation	Write	Avoid
Capture prompt	Move slowly to the left side	Initiate lateral traverse
Finding	Right front door paint differs from the quarter panel	Repaint detected
Uncertainty	Second opinion is unsure, so confidence is lower	Low-confidence anomaly
Next step	Ask the seller if this panel was repainted	Investigate further
Missing angle	Rear-left corner is missing	Coverage incomplete
Upload error	Upload stopped at 62%. Check your connection, then tap Resume.	Something went wrong
Delete	Delete inspection and media	Confirm

Never write "accident-free", "safe to buy", "certified", or repair-cost figures anywhere in the UI. Findings are "possible", with a confidence value and a verification step.

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12. Design tokens (copy-ready)

```
:root {
  /* backgrounds */
  --bg-canvas: #F6F9FC;
  --bg-surface: #FFFFFF;
  --bg-sunken: #EDF2F8;
  --bg-overlay: rgba(255, 255, 255, 0.82);

  /* text */
  --ink-900: #1F2A37;
  --ink-700: #3D4B5C;
  --ink-500: #5F6E80;
  --ink-300: #9AA7B6;

  /* structure */
  --line-subtle: #E1E8F0;
  --line-strong: #C9D4E0;
  --focus-ring: #5B9BE0;

  /* pastels: fill / line / ink */
  --sky-fill: #DCEBFA;      --sky-line: #9CC5EE;
--sky-ink: #1D4E89;
  --mint-fill: #D9F2E6;     --mint-line: #8FD1B0;
--mint-ink: #14513A;
  --butter-fill: #FBF0C9;   --butter-line:
#EBD27A;   --butter-ink: #6B4E00;
  --peach-fill: #FCE0D6;    --peach-line:
#F1A98F;    --peach-ink: #8A3A1F;
  --rose-fill: #F9DCE4;     --rose-line: #E8A0B5;
--rose-ink: #8A2845;
  --lavender-fill: #E8E3F8; --lavender-line:
#B9ABE8; --lavender-ink: #4B3A8C;

  /* shape */
  --radius-chip: 999px;
  --radius-control: 14px;
  --radius-card: 20px;
  --radius-sheet: 28px;
```

```
  --radius-evidence: 12px;

  /* depth */
  --shadow-float: 0 8px 24px rgba(31, 42, 55,
0.08);

  /* type */
  --font-display: "Bricolage Grotesque", system-
ui, sans-serif;
  --font-body: "Figtree", system-ui, -apple-
system, "Segoe UI", sans-serif;
}

body { background: var(--bg-canvas); color: var(-
-ink-700); font-family: var(--font-body); }
h1, h2 { font-family: var(--font-display); color:
var(--ink-900); }
.tabular { font-variant-numeric: tabular-nums; }
:focus-visible { outline: 2px solid var(--focus-
ring); outline-offset: 2px; }

@media (prefers-reduced-motion: reduce) {
  * { animation: none !important; transition:
none !important; }
}
```

Tailwind users: map the same names into `theme.extend.colors` (`sky.fill`, `sky.line`, `sky.ink`, and so on) and `fontFamily.display` / `fontFamily.body`.

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13. Don'ts

- Don't use pastel colours for text. Use the matching ink tone.

- Don't use red or green alone to signal state.
- Don't show a score, a severity or a confidence without the evidence image or measurement it came from.
- Don't hide *uncertain* findings or tuck confidence into a tooltip.
- Don't use dark mode surfaces outside the live camera view (the product direction is light).
- Don't use the same radius and shadow on every element; follow the hierarchy in section 5.
- Don't use all-caps labels, numbered markers on non-sequential content, or arrows on button labels.
- Don't write copy that implies certification, an accident-free guarantee or repair-cost estimates.